

EXPERIMENT 6:

```
use restaurantDB;
```

```
db.restaurants.insertMany([
  {name: "Biryani House", cuisine: "Indian", location: "Jayanagar",
    reviews: [{ user: "Aarav", rating: 5, comment: "Amazing biryani!" },
    { user: "Bhavana", rating: 4, comment: "Great place!" } ]},
  {name: "Burger Joint", cuisine: "American", location: "Koramangala",
    reviews: [{ user: "Chirag", rating: 3, comment: "Average burger" },
    { user: "Devika", rating: 4, comment: "Good value" } ]},
  {name: "Pasta House", cuisine: "Italian", location: "Rajajinagar",
    reviews: [{ user: "Esha", rating: 5, comment: "Delicious pasta!" },
    { user: "Farhan", rating: 4, comment: "Nice ambiance" } ]},
  {name: "Curry Palace", cuisine: "Indian", location: "Jayanagar",
    reviews: [{ user: "Gaurav", rating: 4, comment: "Spicy and tasty!" },
    { user: "Harini", rating: 5, comment: "Best curry in town!" } ]},
  {name: "Taco Stand", cuisine: "Mexican", location: "Jayanagar",
    reviews: [{ user: "Ishaan", rating: 5, comment: "Fantastic tacos!" },
    { user: "Jaya", rating: 4, comment: "Very authentic" } ]} ]]);
```

```
// Aggregation pipeline
```

```
db.restaurants.aggregate([
  { $match: { location: "Jayanagar" } },
  { $unwind: "$reviews" },
  { $group: { _id: "$name",
    averageRating: { $avg: "$reviews.rating" },
    totalReviews: { $sum: 1 } } },
  { $sort: { averageRating: -1 } },
  { $project: { _id: 0,
    restaurant: "$_id", averageRating: 1,
    totalReviews: 1 } },
  { $skip: 1 } ]]).pretty();
```

EXPERIMENT 7: use vacationRentals;

```
db.listingsAndReviews.insertMany([
  {listing_url: "http://www.example.com/listing/123456",
    name: "Beautiful Apartment", address: {street: "123 Main Street",
    suburb: "Central", city: "Metropolis", country: "Wonderland"},
    host: {name: "Alice", picture_url: "http://www.example.com/images/host/host123.jpg"} },
  {listing_url: "http://www.example.com/listing/654321",
    name: "Cozy Cottage", address: {street: "456 Another St",
    suburb: "North", city: "Smallville", country: "Wonderland"},
    host: {name: "Bob", picture_url: "" } },
  {listing_url: "http://www.example.com/listing/789012",
    name: "Modern Condo", address: {street: "789 SideRoad",
    suburb: "East", city: "Gotham", country: "Wonderland"},
    host: {name: "Charlie", picture_url: "http://www.example.com/images/host/host789.jpg"} } ]]);

db.listingsAndReviews.find({ "host.picture_url": { $exists: true, $ne: "" } },
  {listing_url: 1, name: 1, address: 1, "host.picture_url": 1 }).pretty();
```

7B EXP) use ecommerce;

```
db.products.insertMany([
  {product_id: 1, name: "Laptop", category: "Electronics", price: 1200,
    reviews: [{user: "Alice", rating: 5, comment: "Excellent!" },
    {user: "Bob", rating: 4, comment: "Very good" },
    {user: "Charlie", rating: 3, comment: "Average" } ]},
  {product_id: 2, name: "Smartphone", category: "Electronics", price: 800,
    reviews: [{user: "Dave", rating: 4, comment: "Good phone" },
    {user: "Eve", rating: 2, comment: "Not satisfied" },
    {user: "Frank", rating: 5, comment: "Amazing!" } ]},
  {product_id: 3, name: "Headphones", category: "Accessories", price: 150,
    reviews: [{user: "Grace", rating: 5, comment: "Great sound" },
    {user: "Heidi", rating: 3, comment: "Okay" } ]} ]]);

db.products.aggregate([
  { $unwind: "$reviews" },
  { $group: { _id: "$name", totalReviews: { $sum: 1 },
    averageRating: { $avg: "$reviews.rating" },
    comments: { $push: "$reviews.comment" } } },
  { $project: { _id: 0, product: "$_id", totalReviews: 1,
    averageRating: 1, comments: 1 } } ]]).pretty();
```

EXPERIMENT 8:

```
use MoviesDB;
db.createCollection("Movies");
db.Movies.insertMany([
  {title:"Inception",director:"Christopher Nolan",genre:"Science Fiction",year:2010,ratings:{imdb:8.8,rottenTomatoes:87}},
  {title:"The Matrix",director:"Wachowskis",genre:"Science Fiction",year:1999,ratings:{imdb:8.7,rottenTomatoes:87}},
  {title:"The Godfather",director:"Francis Ford Coppola",genre:"Crime",year:1972,ratings:{imdb:9.2,rottenTomatoes:97}},
  {title:"Pulp Fiction",director:"Quentin Tarantino",genre:"Crime",year:1994,ratings:{imdb:8.9,rottenTomatoes:92}},
  {title:"The Shawshank Redemption",director:"Frank Darabont",genre:"Drama",year:1994,ratings:{imdb:9.3,rottenTomatoes:91}},
  {title:"The Dark Knight",director:"Christopher Nolan",genre:"Action",year:2008,ratings:{imdb:9.0,rottenTomatoes:94}},
  {title:"Fight Club",director:"David Fincher",genre:"Drama",year:1999,ratings:{imdb:8.8,rottenTomatoes:79}}]);
db.Movies.createIndex({title:1},{unique:true});

db.Movies.createIndex({title:1},{unique:true});

db.Movies.updateOne({title:"The Godfather"},{$set:{awards:"Best Picture"}});

db.Movies.createIndex({awards:1},{sparse:true});
db.Movies.createIndex({director:1,year:-1});

db.Movies.updateOne({title:"Inception"},{$set:{actors:["Leonardo DiCaprio","Joseph Gordon-Levitt","Elliot Page"]}});
db.Movies.createIndex({actors:1});

db.Movies.find({director:"Christopher Nolan"}).explain("executionStats");
```

EXPERIMENT 9:

```
use CatalogDB;

db.createCollection("CatalogData");

db.CatalogData.insertMany([
  {title:"Node.js Guide",description:"Learn server-side programming with Node.js and JavaScript."},
  {title:"MongoDB Basics",description:"This book introduces MongoDB and NoSQL database design."},
  {title:"Advanced Python",description:"Master advanced concepts in Python for data science and web development."},
  {title:"Java Programming",description:"Comprehensive guide to Java programming for beginners."},
  {title:"Python Data Analysis",description:"Data analysis and visualization with Python and Pandas."},
  {title:"Web Development",description:"Covers HTML, CSS, JavaScript, and modern frameworks."}]);

db.CatalogData.createIndex({title:"text",description:"text"});

db.CatalogData.find({$text:{$search:"Python"}});

db.CatalogData.find({$text:{$search:"-Java"}});

db.CatalogData.find({$text:{$search:"-data science"}});

db.CatalogData.find({$text:{$search:"Python-Pandas"}});

db.CatalogData.getIndexes();
```

EXPERIMENT 10:

use CatalogDB;

```
db.CatalogData.insertMany([ {title:"Node.js Guide",description:"Learn server-side programming with Node.js and JavaScript."}, {title:"MongoDB Basics",description:"This book introduces MongoDB and NoSQL database design."}, {title:"Advanced Python",description:"Master advanced concepts in Python for data science and web development."}, {title:"Java Programming",description:"Comprehensive guide to Java programming for beginners."}, {title:"Python Data Analysis",description:"Data analysis and visualization with Python and Pandas."}, {title:"Web Development",description:"Covers HTML, CSS, JavaScript, and modern frameworks."} ]]);
```

```
db.CatalogData.createIndex({title:"text",description:"text"});
```

```
db.CatalogData.aggregate([ { $match: { $text: { $search: "Python" } } }, { $project: { title: 1, description: 1, score: { $meta: "textScore" } } }, { $sort: { score: -1 } } ]]);
```